ARDOR / INDEPENDENT FUNCTION MODULE

Isolated MIDI input / 3.3 V UART / HAND ASSEMBLY

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

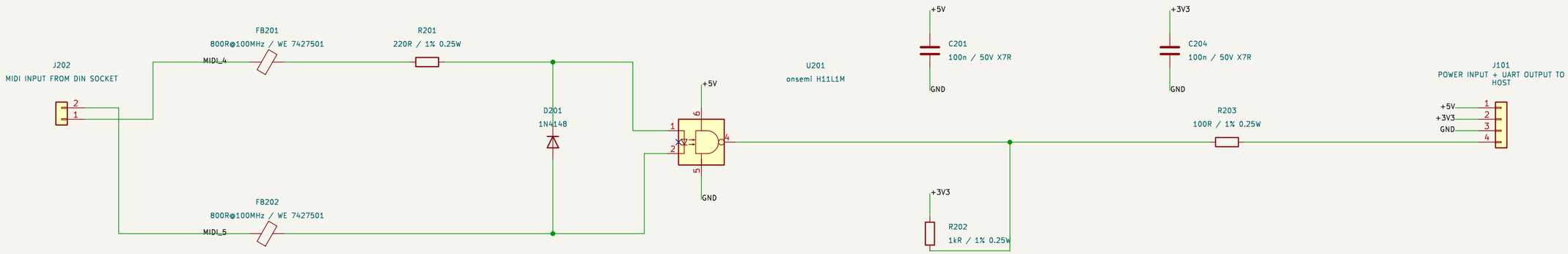
PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J202 / MIDI IN
1: DIN4 IN 2: DIN5 IN

J203 / CASE BOND
1: CASE 2: GND 0V

J101 / PWR IN / UART OUT
1: 5V IN 2: 3V3 IN
3: GND 0V 4: UART OUT

U201.3: NC — LEAVE OPEN



DIN panel socket: contact 4 -> J202.1; contact 5 -> J202.2. Pins 1/2/3 and shell unconnected.
Insulate DIN shell. J203.1 bonds the enclosure; R101 returns ESD to local GND.
Optocoupler LED current -> UART low. The output pull-up is 3.3 V, never 5 V.
Functional ground-loop isolation; transient clamps limit common-mode range. Not a safety isolator.

+3V3 +5V CHASSIS GND

Engineering prototype — bench validation required

Ardor

Sheet: /
File: Ardor_MIDI_THT.kicad_sch

Title: Isolated MIDI input / 3.3 V UART / HAND ASSEMBLY

Size: A3	Date: 2026-10-12	**Rev: T1.1**
KiCad E.D.A. 9.0.2		Id: 1/1

ARDOR / INDEPENDENT FUNCTION MODULE

Passive expression pedal / I2C ADC / HAND ASSEMBLY

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J302 / PASSIVE PEDAL
1: TIP I/O 2: RING I/O
3: GND 0V

JP301 / PEDAL JUMPER
1: TIP 2: RING
3: PEDAL IN 4: EXC OUT
5: RING 6: TIP

JP303 / DATA PULLUP
1: PULLUP 2: DATA I/O

J101 / PWR IN / I2C
1: 3V3 IN 2: GND 0V
3: DATA I/O 4: CLOCK IN

JP302 / ADDR JUMPER
1: LOW GND 2: SELECT
3: HIGH 3V3

JP304 / CLOCK PULLUP
1: PULLUP 2: CLOCK IN

U301.2: UNUSED ALERT – LEAVE OPEN

J302
PASSIVE PEDAL INPUT / EXCITATION OUTPUT
3 GND
2 EXP_RING
1 EXP_TIP

JP301
JUMPER ONLY: PEDAL POLARITY
EXP_TIP 1 2 EXP_RING
3 4
EXP_RING 5 6 EXP_TIP

R301
1kR / 1% 0.25W
+3V3_A — EXP_EXC

D303
BAT85S
+3V3_A

R302
10kR / 1% 0.25W

D304
BAT85S
GND

R303
1MR / 1% 0.25W
GND

C301
100n / 50V X7R
GND

R308
2.2kR / 1% 0.25W
+3V3_A

JP303
JUMPER ONLY: SDA PULLUP
1
SDA 2

R309
2.2kR / 1% 0.25W
+3V3_A

JP304
JUMPER ONLY: SCL PULLUP
1
SCL 2

D301
1.5KE6.8CA
CHASSIS
EXP_TIP

D302
1.5KE6.8CA
CHASSIS
EXP_RING

U301
ADS1115-M
+3V3_A
8 VDD
ALERT/RDY 2
4 AIN0
5 AIN1
6 AIN2
7 AIN3
10 SCL
9 SDA
GND ADDR 1
3 GND
GND
GND

R304
10kR / 1% 0.25W
EXP_EXC

D305
BAT85S
+3V3_A

C302
100n / 50V X7R
GND

R305
33R / 1% 0.25W
SCL

R306
33R / 1% 0.25W
SDA

J101
3V3 POWER INPUT + I2C TO HOST
+3V3 1
GND 2
3
4

JP302
JUMPER ONLY: ADC ADDRESS
GND 1
2
+3V3_A 3

D306
BAT85S
GND

R101
0R / 1% 0.25W
CHASSIS — GND

FB101
800R@100MHz / WE 7427501
+3V3 — +3V3_A

C101
10u / 25V polarized
GND

C102
100n / 50V X7R
GND

C303
100n / 50V X7R
+3V3_A
GND

R307
2.2kR / 1% 0.25W
+3V3_A
GND

Passive 10k–100k pot only. Sleeve = GND. JP301: both shunts 1–3 / 2–4 for tip wiper; 3–5 / 4–6 for ring wiper.
JP302: 1–2 = 0x48; 2–3 = 0x49. Never leave ADDR floating. JP303/304 enable local 2.2k I2C pull-ups.
Remove BOTH pull-up shunts when host already has pull-ups (Raspberry Pi). Start at 100 kHz I2C.
ADS1115: +/-4.096 V, 128 SPS. Read AIN0 and AIN1; normalize AIN0/AIN1, calibrate endpoints and smooth.

+3V3 +3V3_A CHASSIS GND

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_Expression_THT.kicad_sch

Title: Passive expression pedal / I2C ADC / HAND ASSEMBLY

Size: A3	Date: 2026-10-12	**Rev: T1.1**
KiCad E.D.A. 9.0.2		Id: 1/1

ARDOR / INDEPENDENT FUNCTION MODULE

Stereo buffer / mono average / AC outputs / HAND ASSEMBLY

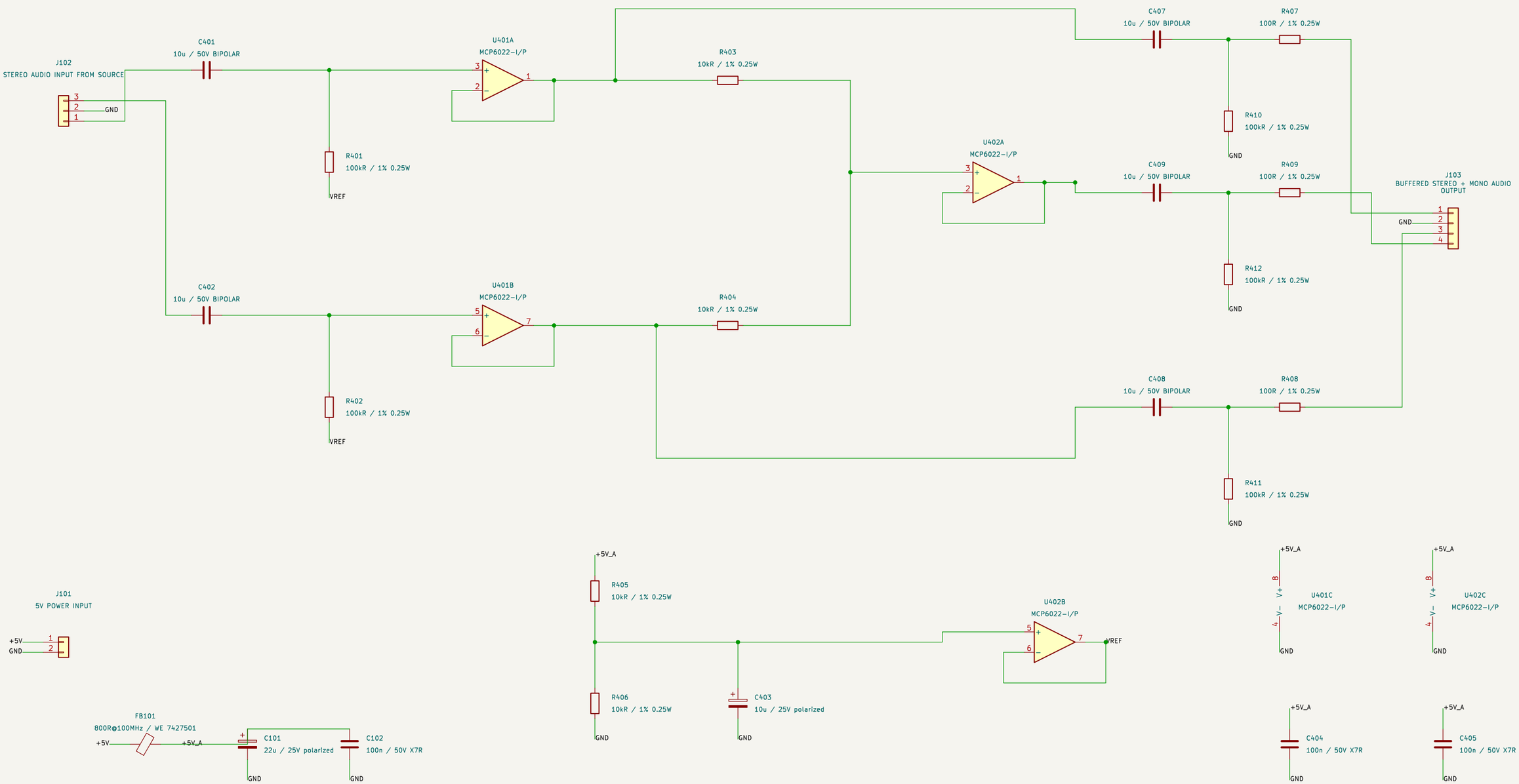
T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J101 / POWER IN
1: 5V IN 2: GND 0V

J102 / AUDIO IN
1: LEFT IN 2: GND 0V
3: RIGHT IN

J103 / AUDIO OUT
1: LEFT OUT 2: GND 0V
3: RIGHT OUT 4: MONO OUT



Line-level input <=1 Vrms/channel. Output loads >=10k. Mono = (L + R)/2, unity buffers.
All three outputs AC coupled; no 2.5 V bias leaves this board. For mono input, feed both channels
or accept −6 dB when using one channel. Harness connectors are internal; add jack-entry ESD protection
for exposed ports. This board is neither an instrument preamp nor an ADC, headphone or power amplifier.

Engineering prototype — bench validation required

Ardor

Sheet: /
File: Ardor_Mixer_THT.kicad_sch

Title: Stereo buffer / mono average / AC outputs / HAND ASSEMBLY

Size: A3	Date: 2026-10-12	**Rev: T1.1**
KiCad E.D.A. 9.0.2		Id: 1/1

ARDOR / INDEPENDENT FUNCTION MODULE

Mono line output / relay mute / internal amp feed / HAND ASSEMBLY

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J101 / PWR + ENABLE IN
1: 5V IN 2: GND 0V
3: ENABLE IN

J102 / AUDIO IN
1: AUDIO IN 2: GND 0V

J503 / LINE OUT
1: LINE OUT 2: GND 0V

J502 / AMP AUDIO OUT
1: AUDIO OUT 2: GND 0V

J102
MONO AUDIO INPUT FROM SOURCE

C401
10u / 50V BIPOLAR

U501A
MCP6022-I/P

C501
47u / 25V polarized

R502
100R / 1% 0.25W

K501
Omron G5V-1 DC5

J503
LINE AUDIO OUTPUT TO TS SOCKET

R401
100kR / 1% 0.25W

VREF

R501
10kR / 1% 0.25W

CHASSIS
D501
1.5KE6.8CA

Q501
2N3904 / E-B-C

R503
470R / 1% 0.25W

D502
1N4148

R504
100kR / 1% 0.25W

J101
5V POWER + ENABLE INPUT

R405
10kR / 1% 0.25W

U501B
MCP6022-I/P

R406
10kR / 1% 0.25W

C403
10u / 25V polarized

FB101
800R@100MHz / WE 7427501

C101
22u / 25V polarized

C102
100n / 50V X7R

U501C
MCP6022-I/P

R101
0R / 1% 0.25W

C404
100n / 50V X7R

J502
AUDIO OUTPUT TO AMP INPUT

C502
10u / 50V BIPOLAR

R505
1kR / 1% 0.25W

R506
100kR / 1% 0.25W

Input <=1 Vrms, input impedance ~100k. LINE load >=10k; internal AMP load >=100k.
Relay OFF: jack tip grounded, source disconnected. Relay ON: unity mono output. No true bypass.
3.3 V active-high enable; leave LOW until audio is stable. 5 V supply budget >=50 mA.
C501 + toward buffer. J502 is a signal feed, not a speaker driver; jack sleeves bond the enclosure.

Engineering prototype — bench validation required

Ardor

Sheet: /
File: Ardor_Line_Out_THT.kicad_sch

Title: Mono line output / relay mute / internal amp feed / HAND ASSEMBLY

Size: A3	Date: 2026-10-12	**Rev: T1.1**
KiCad E.D.A. 9.0.2		Id: 1/1

ARDOR / INDEPENDENT FUNCTION MODULE

Stereo headphones / OPA1656 / relay mute

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J101 / PWR + ENABLE IN
1: 5V IN 2: GND 0V
3: ENABLE IN

J602 / HEADPHONE OUT
1: LEFT OUT 2: RIGHT OUT
3: GND 0V

J102 / AUDIO IN
1: LEFT IN 2: GND 0V
3: RIGHT IN

J102
STEREO AUDIO INPUT FROM SOURCE
C601
10u / 50V BIPOLAR
R611
20k / 1% 0.25W
U601A
OPA1656ID
VREF
R631
10R / 1% 0.25W
C651
470u / 16V
DRIVE_L
R651
10k / 1% 0.25W / UPRIGHT
C661
100p / 50V C0G
R641
1k / 1% 0.25W
GND
CHASSIS
D601
1.5KE6.8CA
HP_L
J602
STEREO HEADPHONE AUDIO OUTPUT
HP_L
HP_R
GND
C602
10u / 50V BIPOLAR
R612
20k / 1% 0.25W
U601B
OPA1656ID
VREF
R632
10R / 1% 0.25W
C652
470u / 16V
DRIVE_R
R652
10k / 1% 0.25W / UPRIGHT
C662
100p / 50V C0G
R642
1k / 1% 0.25W
GND
CHASSIS
D602
1.5KE6.8CA
K601
G5V-2-H1 DC5
HP_R
DRIVE_R
+5V
GND
GND
DRIVE_L
RELAY_LOW
HP_L
HP_R
J101
5V POWER + ENABLE INPUT
GND
+5V
R601
470R / 1% 0.25W
Q601
2N3904 / E-B-C
RELAY_LOW
GND
D603
1N4148
RELAY_LOW
+5V
R602
100k / 1% 0.25W
GND
+5V_A
R621
10k / 1% 0.25W
VREF
R101
0R / WIRE LINK
CHASSIS
GND
FB101
800R@100MHz / WE 7427501
+5V
+5V_A
C101
100u / 25V
C102
100n / 50V X7R
GND
GND
U601C
OPA1656ID
V+
V-
+5V_A
GND
+5V_A
C641
100n / 50V X7R / P2.5mm
R622
10k / 1% 0.25W
C631
47u / 25V
C632
100n / 50V X7R
GND
GND
GND
GND
+5V
+5V_A
CHASSIS
GND

Regulated 5V +/-5% / 150mA, 32-300 ohm headphones; source <=1Vrms.
OPA1656ID: one SOIC-8, 1.27mm pitch; all other components THT.
Inverting gain -10k/20k = -0.5; both inputs held at VREF +2.5V.
10R output isolation BEFORE 470u caps; + toward amplifier. Relay OFF grounds panel.
HP_ENABLE: 3.3V HIGH sources ~5mA; wait >=5s after stable supply/audio.
No onboard timer. Check DC/transients, oscillation, noise and clipping on dummy loads.

Engineering prototype — bench validation required

Ardor

Sheet: /
File: Ardor_Headphones_THT.kicad_sch

Title: Stereo headphones / OPA1656 / relay mute

Size: A3	Date: 2026-10-12	**Rev: T1.1**
KiCad E.D.A. 9.0.2		Id: 1/1